



Fig. 4: 10A, 12V 18650 BMS module

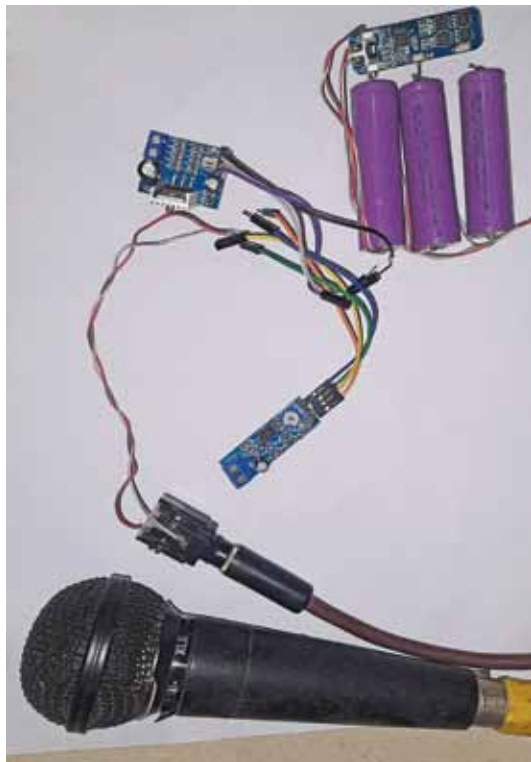


Fig. 5: Author's prototype

output power amplifier. A 6.5mm mono microphone female socket can be used for connecting the microphone to the preamplifier.

18650 BMS module. To make the system portable, the 18650 model lithium-ion 2500mAh, 3.7V rechargeable batteries are used. The batteries have a high energy density

and low self-discharge. Lithium-ion batteries can be recharged hundreds of times and are highly stable. This provides better power efficiency as a single cell has longer charge retention than other rechargeable battery types.

However, deep discharge makes the voltage of lithium-ion cells fall below a certain level where these cannot be recharged further and become useless. To avoid this, we use the 3S 10A, 12V 18650 lithium battery charger board protection module (Fig. 4). Here, 3S stands for the three 18650 batteries' series combination, 10A stands for the maximum discharge current limit, and 12V stands for the operating voltage.

The BMS has an overcharge detection of 4.25V and overdischarge detection of 2.5V per cell. The batteries and the BMS module are connected as shown in the connection diagram. A 12V, 1A SMPS

charger can be connected for charging and powering up the circuit. Choose a suitable box to fit all the modules, the speaker, and the batteries. The author's prototype is shown in Fig. 5.

Precautions. Check the polarities of the modules before making connections. Never short a battery. Lithium-ion batteries are highly sensitive to high temperature, so avoid exposure to high temperatures. Maintain some distance between the mic and the modules to avoid feedback echo noise. **EFY**

PARTS LIST

1. LM386 audio amplifier module
2. TDA2030A audio amplifier module
3. 3S 10A, 12V 18650 lithium battery charger board protection module
4. 18650 model lithium-ion 2500mAh, 3.7V rechargeable batteries (3 nos.)
5. 8-ohm, 15W speaker
6. 6.5mm mono microphone female socket
7. 12V, 1A SMPS charger
8. Connecting wires
9. Dynamic microphone



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